

Off-policy Reinforcement Learning with High Dimensional Reward

Summary

December 23, 2025

1 Problem Setup

Traditional reinforcement learning assumes scalar rewards, but many real-world problems involve high-dimensional or even infinite-dimensional reward spaces. For example, in robotics, rewards might include position, velocity, force, and other continuous measurements. This paper extends distributional reinforcement learning to handle high-dimensional reward spaces, providing theoretical foundations and practical algorithms for these complex problems.

2 Thesis

The central thesis is that distributional reinforcement learning can be generalized to high-dimensional reward spaces, including infinite-dimensional Banach spaces. The authors argue that many real-world problems naturally involve high-dimensional rewards, and that learning the full return distribution (rather than just expectations) is crucial for optimal decision-making in these settings.

3 Key Concepts

High-Dimensional Return Distribution: The probability distribution over cumulative future rewards in high-dimensional spaces, where each dimension represents a different aspect of the reward signal.

Banach Space Rewards: A mathematical framework for handling infinite-dimensional reward spaces using the theory of Banach spaces (complete normed vector spaces).

Distributional Bellman Operator in High Dimensions: A generalization of the distributional Bellman operator that operates on probability distributions over high-dimensional reward spaces.

Wasserstein Distance in High Dimensions: A metric for comparing probability distributions in high-dimensional spaces that preserves the contraction properties needed for convergence.

Utility Function: A function that maps high-dimensional return distributions to scalar values for decision-making, allowing for complex preferences over risk and trade-offs.

4 Methods

The authors develop several key methodological contributions:

1. **Theoretical Framework:** They establish the mathematical foundations for distributional RL in high-dimensional spaces, proving contraction properties of the Bellman operator in Banach spaces.
2. **Dimensionality Reduction:** They show how high-dimensional reward distributions can be effectively approximated using lower-dimensional Euclidean spaces while preserving essential information.

3. **Off-policy Algorithm:** They develop an off-policy RL algorithm that can learn from high-dimensional reward signals without requiring on-policy exploration.
4. **Practical Implementation:** They provide methods for implementing high-dimensional distributional RL using table-based approaches and discuss extensions to deep learning methods.

The key insight is that the contraction properties that make distributional RL work in scalar settings extend to high-dimensional spaces, enabling stable learning even with complex reward structures.

5 Results

The authors demonstrate their approach on several challenging problems:

- **Theoretical Convergence:** They prove that the distributional Bellman operator remains a contraction in high-dimensional spaces, ensuring convergence of the learning algorithm.
- **Dimensionality Reduction:** They show that high-dimensional reward distributions can be effectively approximated using much lower-dimensional representations without significant loss of performance.
- **Off-policy Learning:** Their algorithm successfully learns optimal policies from high-dimensional reward signals using off-policy data, demonstrating sample efficiency.
- **Complex Utility Functions:** The approach can handle complex utility functions that capture sophisticated preferences over risk and trade-offs between different reward dimensions.
- **Simulation Results:** In simulated environments with high-dimensional rewards, the algorithm achieves performance comparable to or better than traditional RL methods that would require scalar reward engineering.

The results show that high-dimensional distributional RL can solve problems that are intractable for conventional reinforcement learning approaches.

6 Significance

This work extends distributional reinforcement learning to handle complex, high-dimensional reward structures with several important implications:

- **Real-World Applications:** Provides a framework for problems with naturally high-dimensional rewards like robotics, finance, and healthcare where multiple continuous measurements must be considered simultaneously.
- **Theoretical Foundation:** Establishes rigorous mathematical foundations for distributional RL in infinite-dimensional spaces, extending the theory beyond the scalar case.
- **No Reward Engineering:** Eliminates the need to manually engineer scalar reward functions from high-dimensional measurements, allowing the agent to learn directly from the raw high-dimensional signals.
- **Rich Decision Making:** Enables more sophisticated decision-making by allowing complex utility functions that capture nuanced preferences over different aspects of the reward.

- **Future Directions:** Opens up research avenues for combining high-dimensional distributional RL with deep learning, particularly transformer-based approaches for handling sequential dependencies.

The paper demonstrates that the benefits of distributional RL extend beyond scalar rewards to complex, high-dimensional settings, providing a foundation for handling the full complexity of real-world decision-making problems.